

INNOVATION CODES:

```
#include <Wire.h>
```

```
#include <Adafruit_MPU6050.h>
```

```
#include <Adafruit_Sensor.h>
```

```
#include <TinyGPS++.h>
```

```
Adafruit_MPU6050 mpu;
```

```
TinyGPSPlus gps;
```

```
HardwareSerial sim800(2);
```

```
HardwareSerial gpsSerial(1);
```

```
#define OVERRIDE 12
```

```
#define BUZZER 14
```

```
#define FOOTPEG1 32
```

```
#define FOOTPEG2 33
```

```
String riderName = "RIDER NAME";
```

```
String bloodGroup = "O+";
```

```
String phoneNumber = "+91XXXXXXXXXX"; // <<< CHANGE NUMBER
```

```
float baseAngle = 102.84; // Upright normal angle
```

```
float fallThreshold = 45.0; // degrees tilt to detect fall
```

```
void setup() {  
  Serial.begin(115200);  
  
  pinMode(OVERRIDE, INPUT_PULLUP);  
  pinMode(BUZZER, OUTPUT);  
  pinMode(FOOTPEG1, INPUT_PULLUP);  
  pinMode(FOOTPEG2, INPUT_PULLUP);  
  
  Wire.begin(21, 22);  
  mpu.begin();  
  
  sim800.begin(9600, SERIAL_8N1, 16, 17);  
  gpsSerial.begin(9600, SERIAL_8N1, 4, 15);  
  
  delay(2000);  
  
  Serial.println("SYSTEM INITIALIZED");  
  Serial.println("=====");  
}  
  
void loop() {  
  
  unsigned long cycleStart = millis();  
  int fallCount = 0;  
  int totalReadings = 0;
```

```
float initLat = 0, initLng = 0;

Serial.println("\nReading sensors for 10 seconds...");

while (millis() - cycleStart < 10000) {

    readGPS();

    float angle = getTiltAngle();
    float tilt = abs(angle - baseAngle);

    int foot1 = digitalRead(FOOTPEG1);
    int foot2 = digitalRead(FOOTPEG2);

    if (gps.location.isValid()) {
        initLat = gps.location.lat();
        initLng = gps.location.lng();
    }

    Serial.print("Angle: ");
    Serial.print(angle);
    Serial.print(" Tilt: ");
    Serial.print(tilt);
    Serial.print(" Foot1: ");
    Serial.print(foot1);
    Serial.print(" Foot2: ");
```

```
Serial.println(foot2);
```

```
if (tilt > fallThreshold) {  
    fallCount++;  
}
```

```
totalReadings++;  
delay(500);  
}
```

```
float fallRatio = (float)fallCount / totalReadings;
```

```
Serial.print("Fall Ratio: ");  
Serial.println(fallRatio);
```

```
if (fallRatio > 0.6) {  
    triggerAlert("FALL DETECTED", initLat, initLng);  
}
```

```
Serial.println("Waiting 20 seconds before next cycle...");  
delay(20000); // completes 30 sec cycle  
}
```

```
float getTiltAngle() {  
    sensors_event_t a, g, temp;  
    mpu.getEvent(&a, &g, &temp);
```

```

float angle = atan2(a.acceleration.y, a.acceleration.z) * 180 / PI;
return angle;
}

void triggerAlert(String type, float initLat, float initLng) {

    Serial.println("\n=====");
    Serial.println("ALERT DETECTED: " + type);
    Serial.println("Press override button within 10 seconds to cancel.");
    Serial.println("=====");

    for (int i = 0; i < 3; i++) {
        tone(BUZZER, 2000);
        delay(300);
        noTone(BUZZER);
        delay(300);
    }

    unsigned long startWait = millis();

    while (millis() - startWait < 10000) {
        if (digitalRead(OVERRIDE) == LOW) {
            delay(200);
            if (digitalRead(OVERRIDE) == LOW) {
                Serial.println("ALERT CANCELLED");
                return;
            }
        }
    }
}

```

```
    }  
    }  
}
```

```
    sendSMS(type, initLat, initLng);  
}
```

```
void sendSMS(String status, float initLat, float initLng) {
```

```
    Serial.println("\n----- SMS PREVIEW -----");
```

```
    readGPS();
```

```
    float finalLat = gps.location.lat();
```

```
    float finalLng = gps.location.lng();
```

```
    String message = "";
```

```
    message += "Rider: " + riderName + "\n";
```

```
    message += "Blood Group: " + bloodGroup + "\n";
```

```
    message += status + "\n\n";
```

```
    message += "Initial Location:\n";
```

```
    if (initLat != 0) {
```

```
        message += "https://maps.google.com/?q=" + String(initLat,6) + "," +  
String(initLng,6) + "\n";
```

```
    } else {
```

```
        message += "GPS Not Fixed\n";
```

```
}
```

```
message += "\nFinal Location:\n";
```

```
if (gps.location.isValid()) {
```

```
    message += "https://maps.google.com/?q=" + String(finalLat,6) + "," +  
String(finalLng,6) + "\n";
```

```
} else {
```

```
    message += "GPS Not Fixed\n";
```

```
}
```

```
message += "\nDate & Time:\n";
```

```
if (gps.date.isValid() && gps.time.isValid()) {
```

```
    message += String(gps.date.day()) + "/" +
```

```
        String(gps.date.month()) + "/" +
```

```
        String(gps.date.year()) + " ";
```

```
    message += String(gps.time.hour()) + ":" +
```

```
        String(gps.time.minute());
```

```
} else {
```

```
    message += "Not Available";
```

```
}
```

```
Serial.println(message);
```

```
Serial.println("-----\n");
```

```
Serial.print("Sending SMS to: ");
```

```
Serial.println(phoneNumber);
```

```
sim800.println("AT+CMGF=1");  
delay(1000);
```

```
sim800.print("AT+CMGS=\"");  
sim800.print(phoneNumber);  
sim800.println("\");  
delay(1000);
```

```
sim800.print(message);  
delay(500);  
sim800.write(26); // CTRL+Z  
delay(5000);
```

```
Serial.println("SMS SENT SUCCESSFULLY");  
}
```

```
void readGPS() {  
    while (gpsSerial.available()) {  
        gps.encode(gpsSerial.read());  
    }  
}
```

Code output (Execution):

```
SYSTEM INITIALIZED
=====

Reading sensors for 10 seconds...
Angle: -53.52  Tilt: 156.36  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Angle: -92.25  Tilt: 195.09  Foot1: 0  Foot2: 0
Fall Ratio: 1.00

=====
ALERT DETECTED: FALL DETECTED
Press override button within 10 seconds to cancel.
=====
```

----- SMS PREVIEW -----

Rider: RIDER NAME

Blood Group: O+

FALL DETECTED

Initial Location:

GPS Not Fixed

Final Location:

GPS Not Fixed

Date & Time:

Not Available

Sending SMS to: +91XXXXXXXXXX

SMS SENT SUCCESSFULLY

Waiting 20 seconds before next cycle...

```
Reading sensors for 10 seconds...
```

[illegible]

Fall Ratio: 1.00

=====

ALERT DETECTED: FALL DETECTED

Press override button within 10 seconds to cancel.

=====

ALERT CANCELLED

```
Waiting 20 seconds before next cycle...
```